HERIOT-WATT UNIVERSITY

G11WT RESERVOIR SIMULATION

COURSE WORK 2024/2025

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H00481039

PART A CALCULATION OF STOIIP AND POROSITY

STOIIP is calculated using the material balance equation. The assumed Sw(int) of 0.28 is not used in the calculations because it includes the water leg where there is almost 100% water and saturation of oil is negligible. The Sw(int) used in the equation is taken from the relative permeability tables where the highest capillary pressure is respective to the immobile water. The Sw(int) of the five layer is averaged to obtained to get one value to be used in the material balance equation. Two pressure values above the bubble point pressure are selected because above the bubble point, the fluid is a single phase and the pressure changes are influenced by the total compressibility of the reservoir and production data rather than below the Bp, where the material balance equation cannot be applied because there would be free gas present and the change in pressure would not be accounted for the compressibility of the fluids and the formation.

Material balance equation assumes the reservoir to have the same properties as a single tank model with homogeneous properties across its geometry such as total compressibility, porosity permeability average pressure. however, as production progresses, the reservoir properties and pressure change. Material balance model does not accurately account for the reservoir heterogeneity of the reservoir thus having inconsistencies int the calculations of STOIIP. PVT properties such as Bo are quite sensitive to pressure change and if not accounted for could result into different values for STOIIP. any errors made during the measurements of static BHP and cumulative oil production values could also result in the fluctuations of STOOIP values.

Arithmetic average method was used to obtain one value as a representation for STOIIP. This method provides a rough estimate of STOIIP and minimizes the errors and inaccuracies made by the material balance model during the STOIIP calculations. Porosity value calculated was 0.32.

The FPR and the static BHP show a good match. This shows that the STOIIP and porosity calculations were done correctly and that the material balance equation model showed a good representation of the reservoir properties, resulting into a proper alignment of the plot between the calculated FPR and the measured static BHP.
The match of FPR vs static BHP is good, compared to the match of the BHP vs BHPH. The good match between FPR and BHP show that a uniform porosity was used by the model to simulate field average pressure and have a good representation of the reservoir pressure conditions and properties, therefore aligning properly with the static BHP. BHP values are dependant of the flow rate conditions of fluids controlled by near wellbore conditions such as skin, wellbore storage. the flow-rate conditions would not be the same as measured in the well were the historical BHPH values were recorded, thus not matching with the BHP values. The perfect match between FOPR AND FOPRH is due to the use of historical data to calculate for the reservoir parameters such as average pressure, BHP to create a good history match between calculated and historical values.

The matches of the calculated and historical data simulated by the model can be made better by adjusting permeability near the wellbore environment could increase flow rate and account for skin and wellbore storage that would cause restrictions of flow giving different bhp than the historical values. Refinement of the grid section by forming smaller grid sections limits numerical errors in the simulation thus providing a good model for history matching.
History matching does not provide an accurate prediction because the reservoir conditions and production rates are not consistent.

PART B HISTORY MATCHING.

The absolute permeability of layer 2 was adjusted with different k values i.e 2000md, 3000md, 4000md, 4800md. the model with 4800 md in layer 2 was chosen because it gave the best match of the bhp of the wells and the FWPR with the historical data.

The match can be improved further by adjusting the relative permeability of fluid and vertical permeability (kv/kh).

The parameters that cannot be changed are porosity and the geometry of the reservoir to prevent errors from being made when the model is doing calculations and simulations.

PART C

A total of 12 simulations were conducted in trying to predict the future production performance of the field with enhanced recovery methods. The cases in this simulation were drilling new producer wells and side tracking the existing D10 well. The intention was to increase drainage positions in the undrained areas reservoir resulting into an increased amount of oil production of about 3.2MM STB and a significant increase with the oil recovery efficiency of about 53%. Gas and water injection is done to maintain the reservoir pressure and increase the displacement factor of oil towards the producer. These sensitivities however did not result into improved oil recovery as the total production reduced to an average of 2.6 MM STB and oil recovery factor of 45%, lower than just drilling new wells. Polymer injection showed the highest oil sweep efficiency thus producing more than in the other cases with the oil recovery factor increasing to 60%. The water cut across all the cases remained at an average of 86%.
The best recovery method among the all the cases was polymer flooding combined with a new horizontal well. The polymer introduced in the water injected increases the viscosity of the water, improving mobility ratio, resulting into a more uniform displacement of oil without viscous fingering. The horizontal well increases the surface area contact with the reservoir enhancing the oil production. Amidst utilising the budget of the two wells, this method justifies the economy limits constructed by producing at a high efficiency.

The optimum model provides the best strategy compared to the nothing done case because polymer flooding increases the displacement of oil by the water uniformly and the horizontal well increases the contact surface to the reservoir, thus producing more oil. Whereas the nothing done case relies on the previous completions and natural drive mechanisms of the reservoir during production resulting onto a lower recovery for short period regardless of the economic constraints of 400stb/day and 98% water cut established in the simulation model. The results table shows that the optimum model, case 12 has an increased oil recovery factor by 33.6% and an increased total field production by 19.9 %.

COMPARISON OF BEST CASES BETWEEN POLYMER FLOODING VERSES GAS INJECTION.

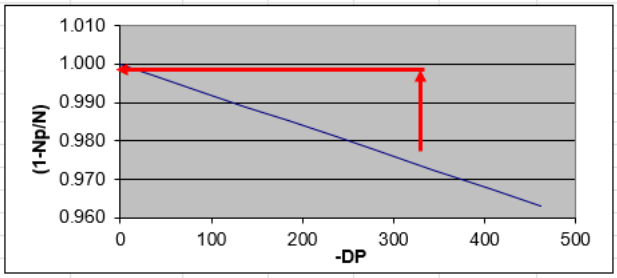
Polymer flooding: polymer injection provides a more uniform sweep of oil preventing viscous fingering and the horizontal well increases the surface area contact with the reservoir resulting into an increase in oil production. This case has an oil recovery factor of 64% compared to case 10 with polymer flooding alone and having an oil efficiency of 60%. This shows that the horizontal well in case12 is an added advantage to the polymer flooding making it more effective as a recovery method. **Gas injection**: For case 9, the horizontal well is completed in layer 2 that has a high permeability, and the gas well is completed to layer 5. Since gas is less dense, its rises upwards and displaces the oil in the upper layers thus having a good displacement efficiency, relatively good oil recovery of 57% and a total field production of 3.9MM STB.

A	B	C	D	E	F	G
P	DP	- DP	Bo	(1-Np/N)	Np	**N**
5102	0	0	1.030686	1.000000	0	**0**
5063	-39	39	1.033670	0.996885	17149	**5,505,667**
5020	-82	82	1.036981	0.993452	44013	**6,722,089**
4964	-138	138	1.041323	0.988984	73843	**6,703,468**
4918	-184	184	1.044918	0.985316	103673	**7,060,342**
4865	-237	237	1.049091	0.981092	132542	**7,009,842**
4818	-284	284	1.052819	0.977348	164200	**7,248,846**
4759	-343	343	1.057537	0.972651	190876	**6,979,281**
4696	-406	406	1.062621	0.967639	222151	**6,864,748**
4640	-462	462	1.067182	0.963186	251592	**6,834,230**

H	I	J	K	L
	Bo(init)	Sw(init)	Cw	Cf
	1.030686	0.21	3.0E-06	4.0E-06
PVT data	P	Bo	Slope	Const
	5111	1.03	7.182E-05	0.603808
	3100	1.21		
	AVERAGE STOIIP		**6,769,835**	

average Swc of layers	
layer 1	0.25
layer 2	0.15
layer 3	0.15
layer 4	0.25
layer 5	0.25
avg	0.21

porosity calculation	
Bo(int)	1.030686
So	0.79
bulk vl	37,279,000
NTG	0.75
N	6769835
Ø	0.315902
POROSITY	**0.32**



Note ECLIPSE 100 linearly interpolates the **reciprocals** of B_o and $(B_o\mu_o)$ between data points, rather than the values themselves. This should be taken into account when comparing the results of ECLIPSE 100 with those of other simulators, by ensuring the data points are not distributed too sparsely.

Figure 1 SHOWING THE CALCULATION OF STOOIP AND POROSITY

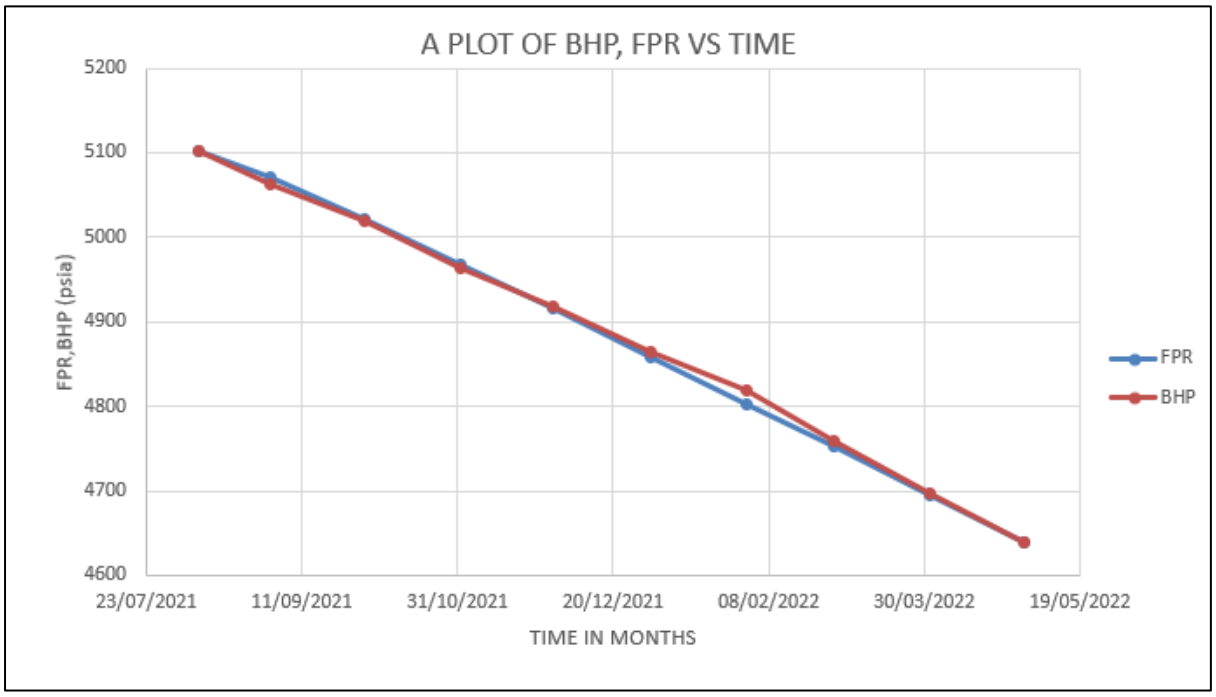


FIGURE 2 plot of Field Average Pressure (FPR) from the model and the static BHP data

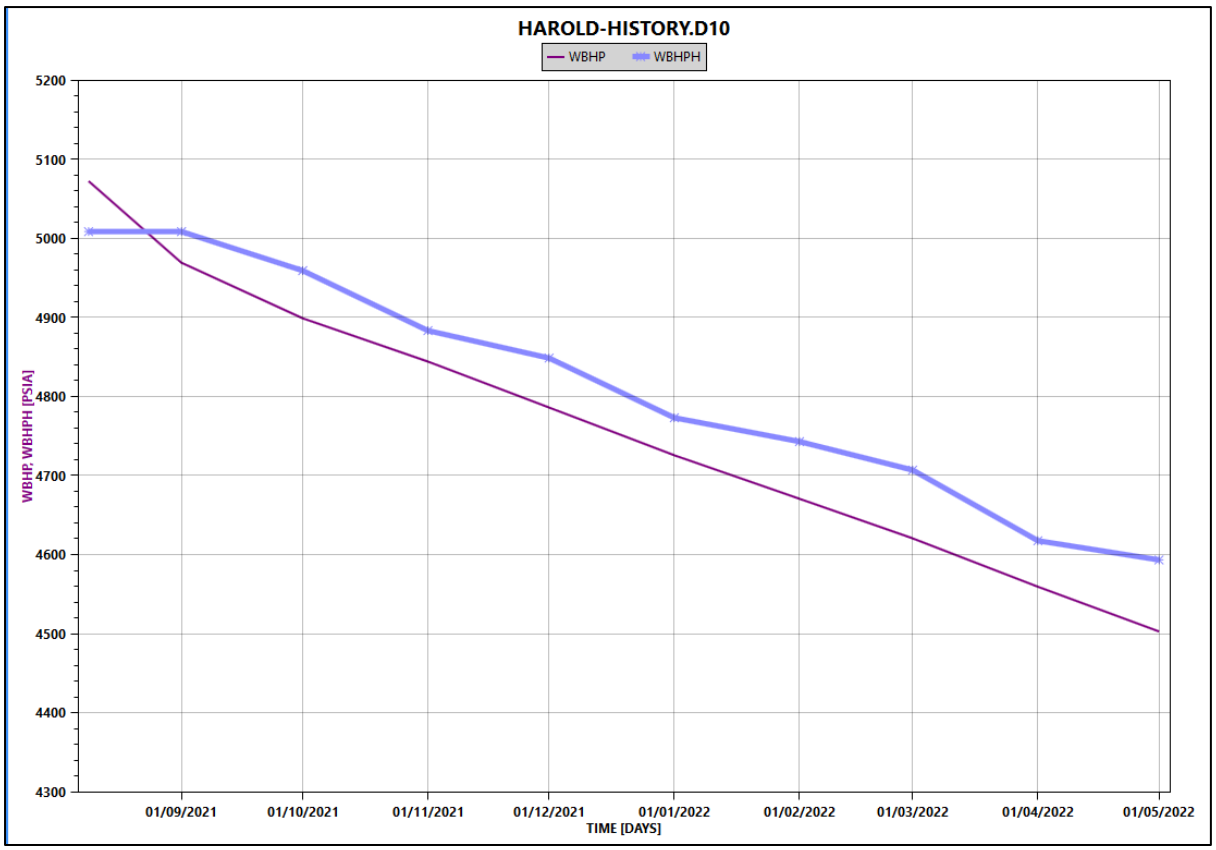


FIGURE 3 PLOT OF WBHP AND WBHPH AGAINST TIME

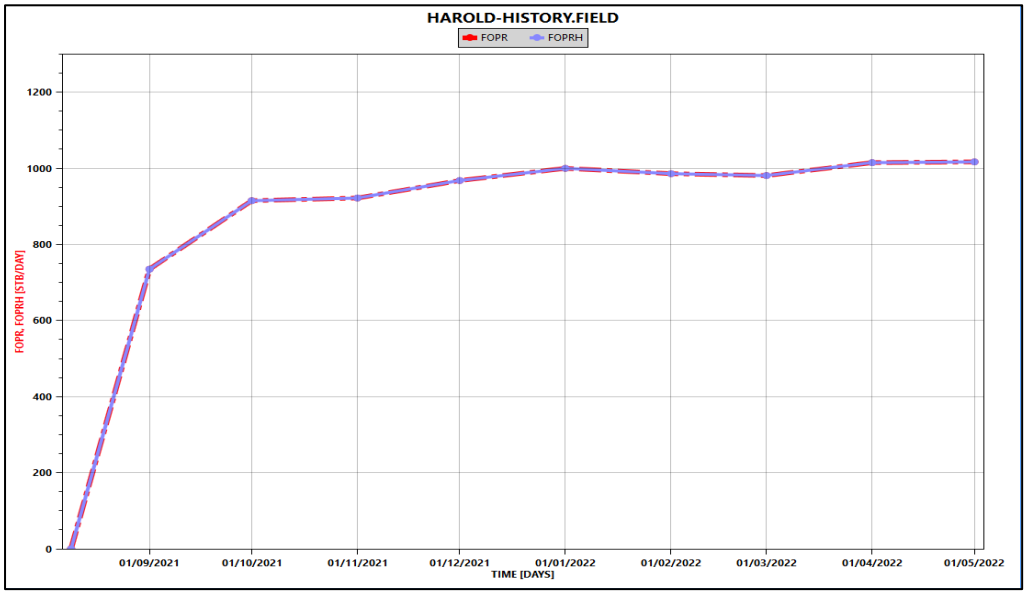


FIGURE 4: PLOT FOPR AND FOPRH AGAINST TIME

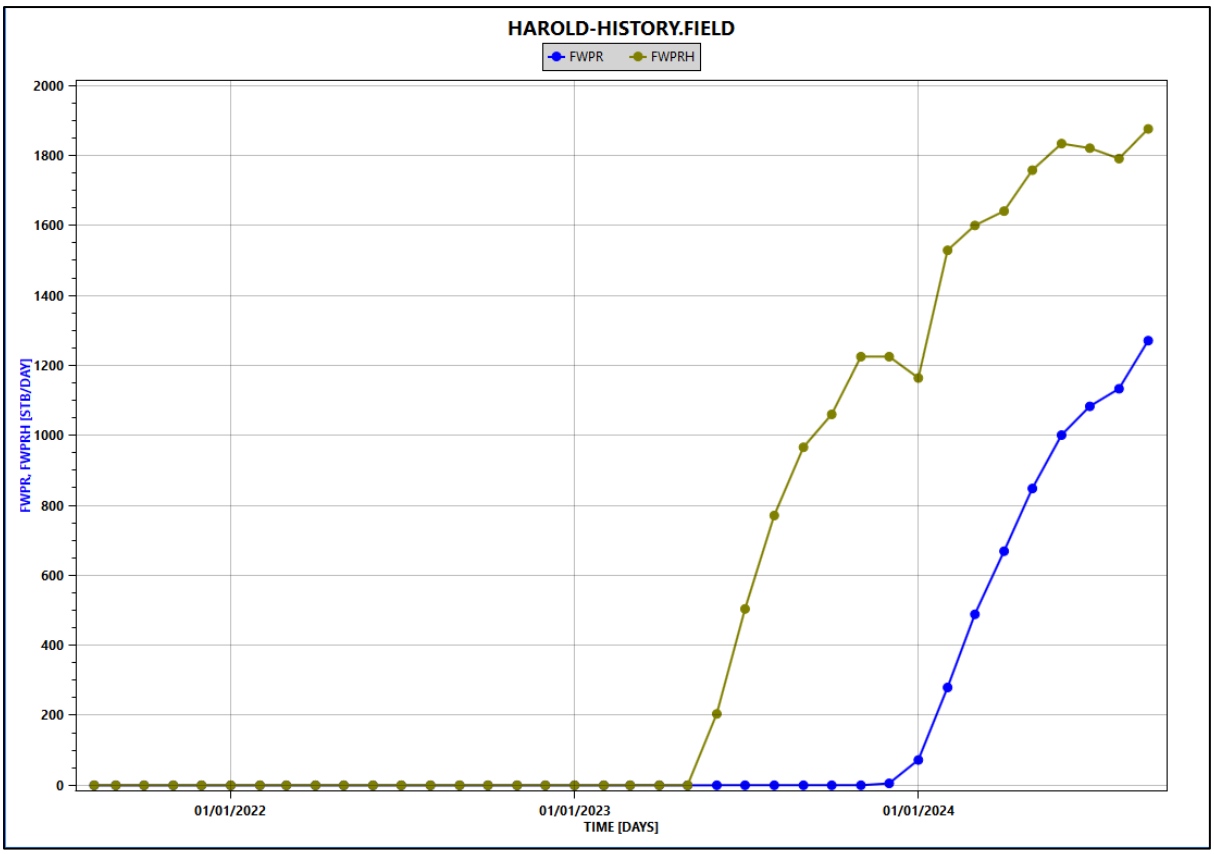


Figure 5 plot calculated and historical water production rates.

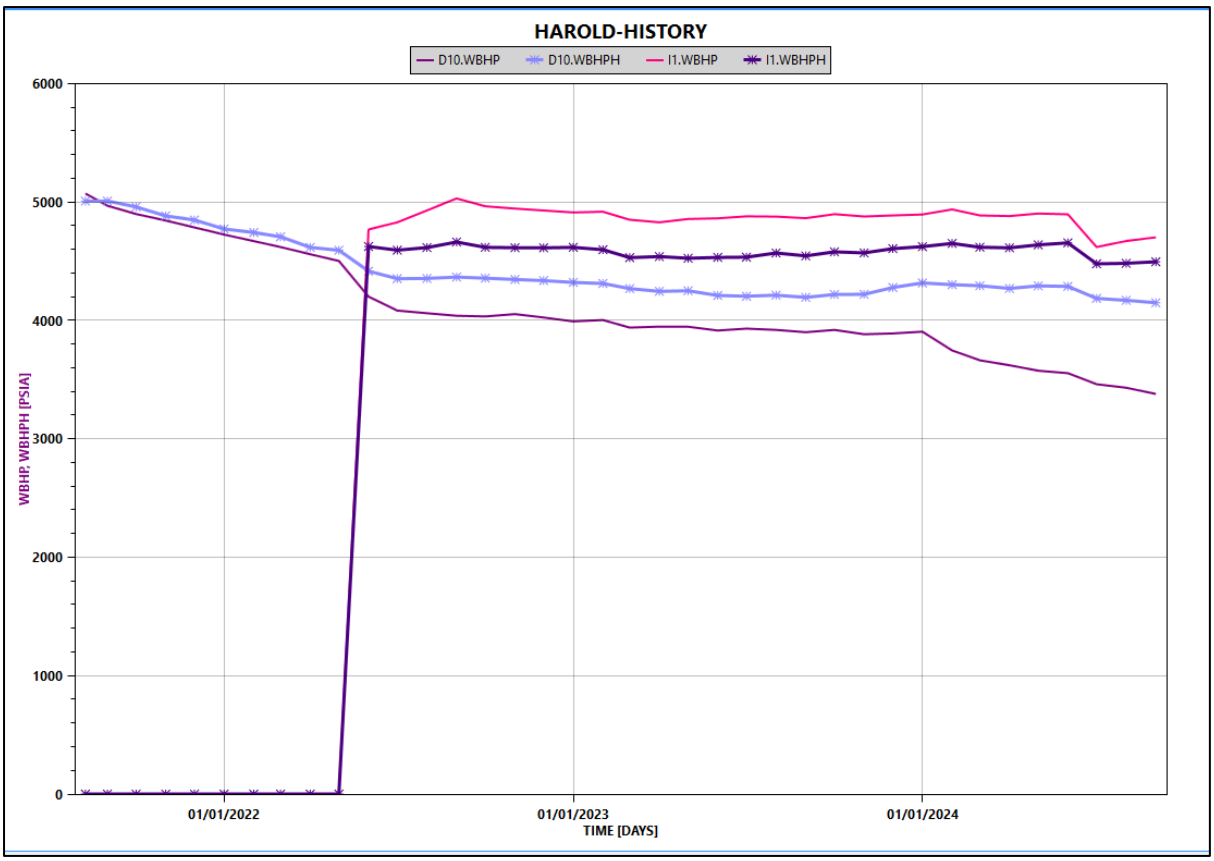


Figure 6 plot the calculated and historical BHP values at the injector and producer wells.

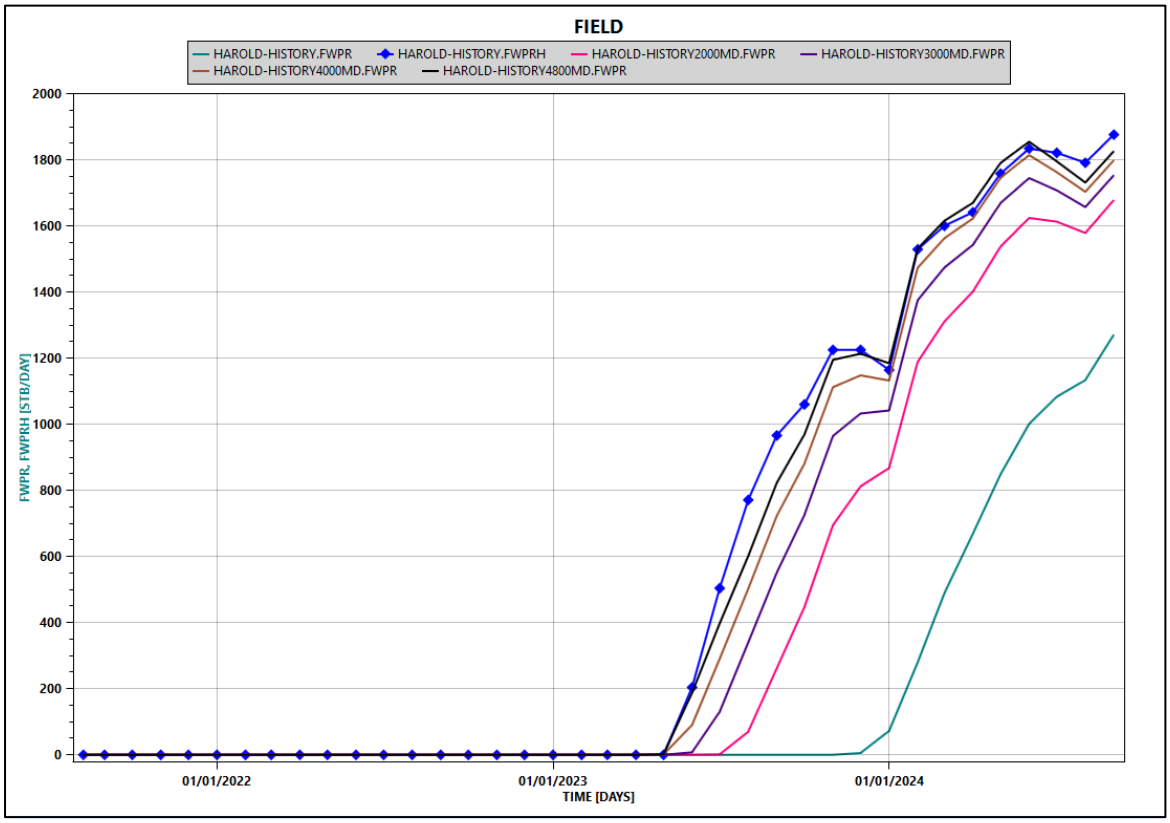


FIGURE 7 SHOWING PLOT CALCULATED AND HISTORICAL WATER FLOW RATES SIMULATIONS

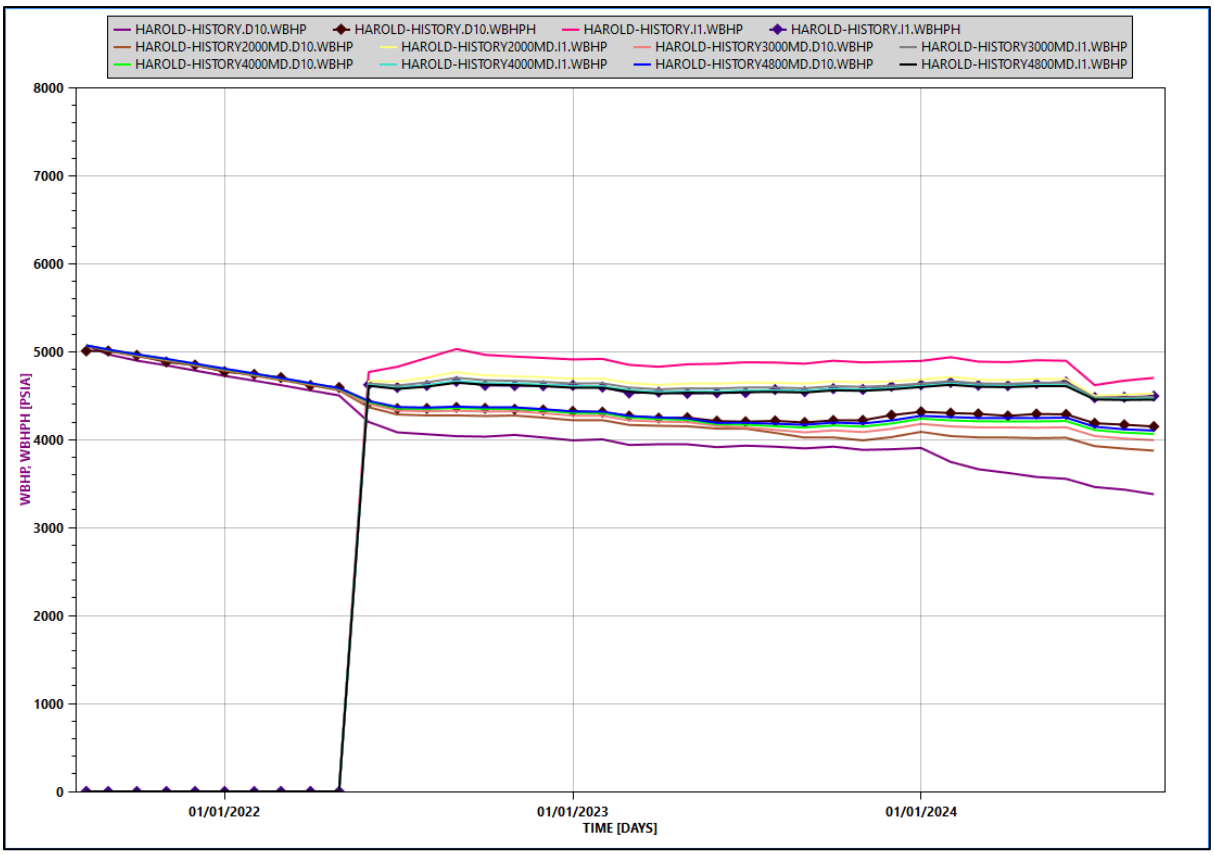


Figure 8 showing plot of the BHPs for producer and injector well simulations

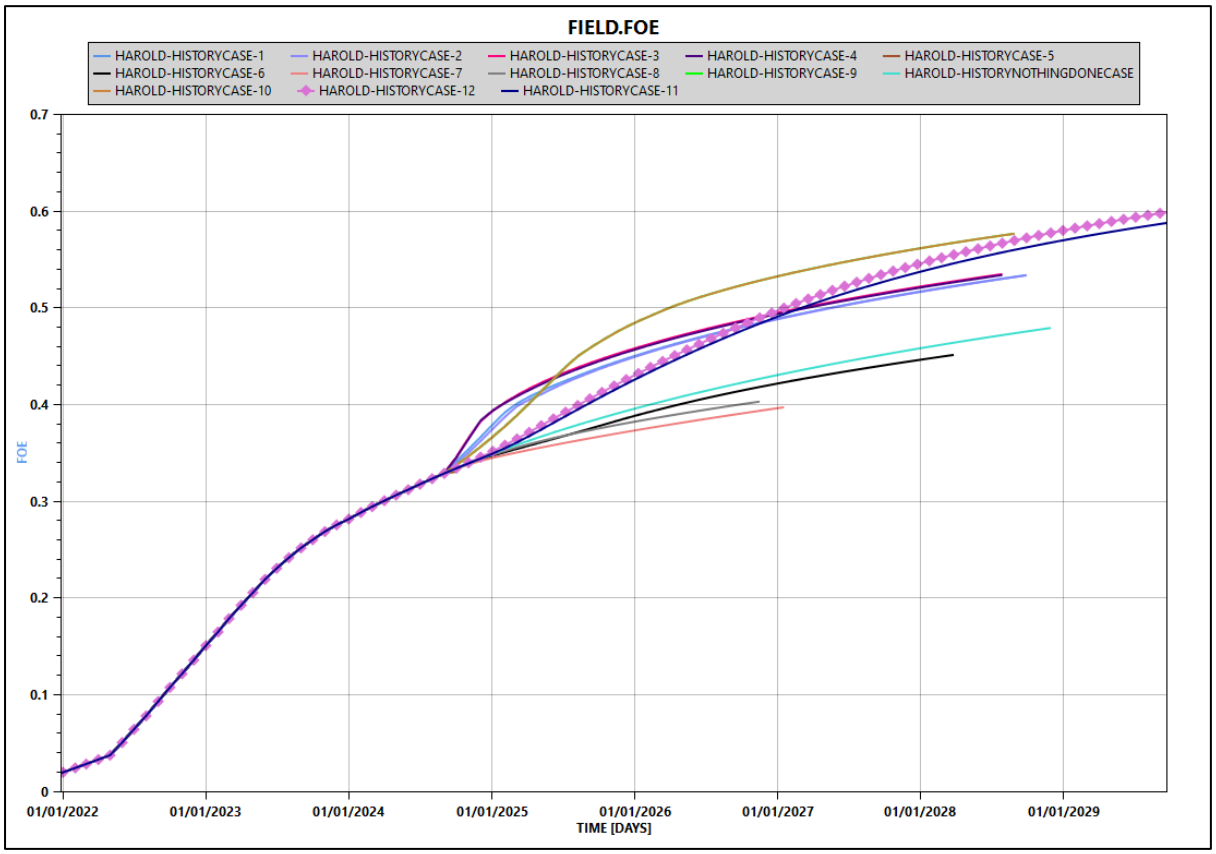


FIGURE 9 SHOWING RECOVERY FACTOR FOR ALL THE CASES DONE

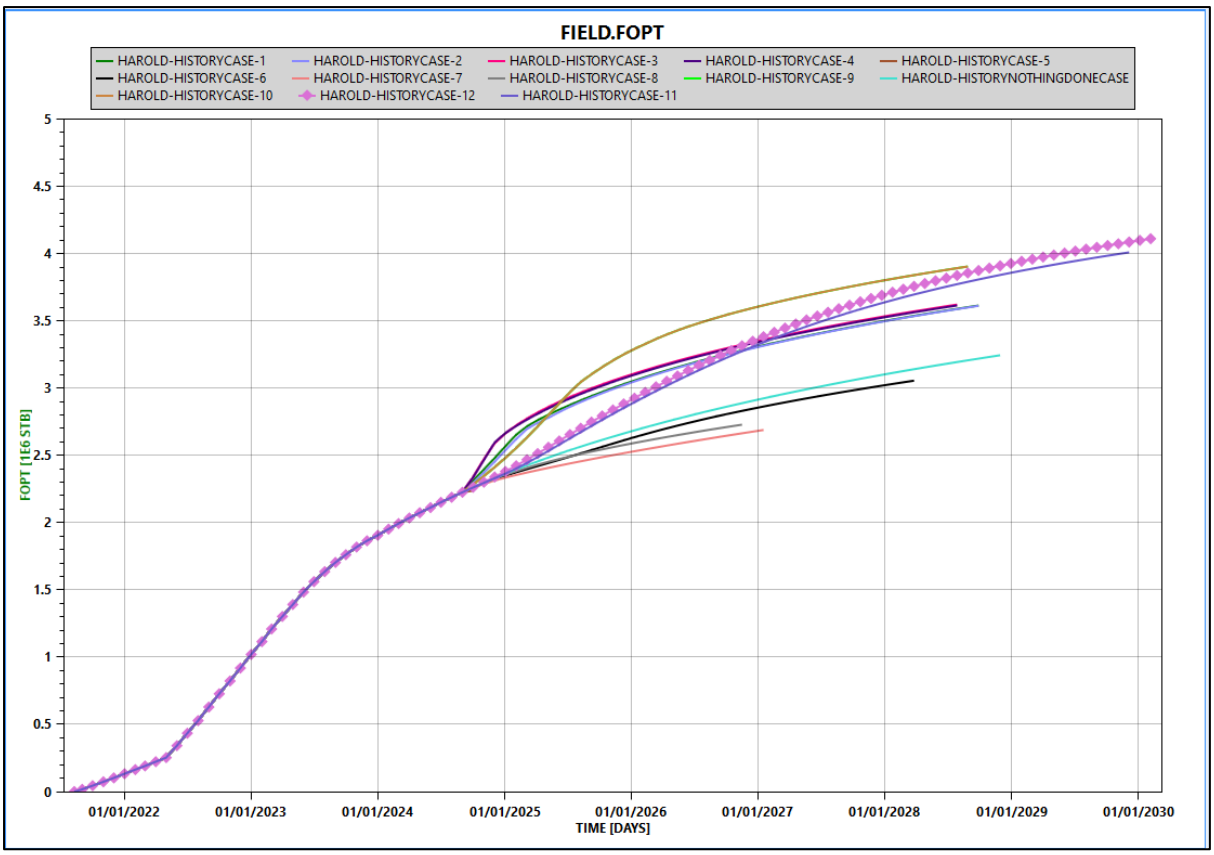


FIGURE 10. SHOWING FOPT FOR ALL THE CASES DONE

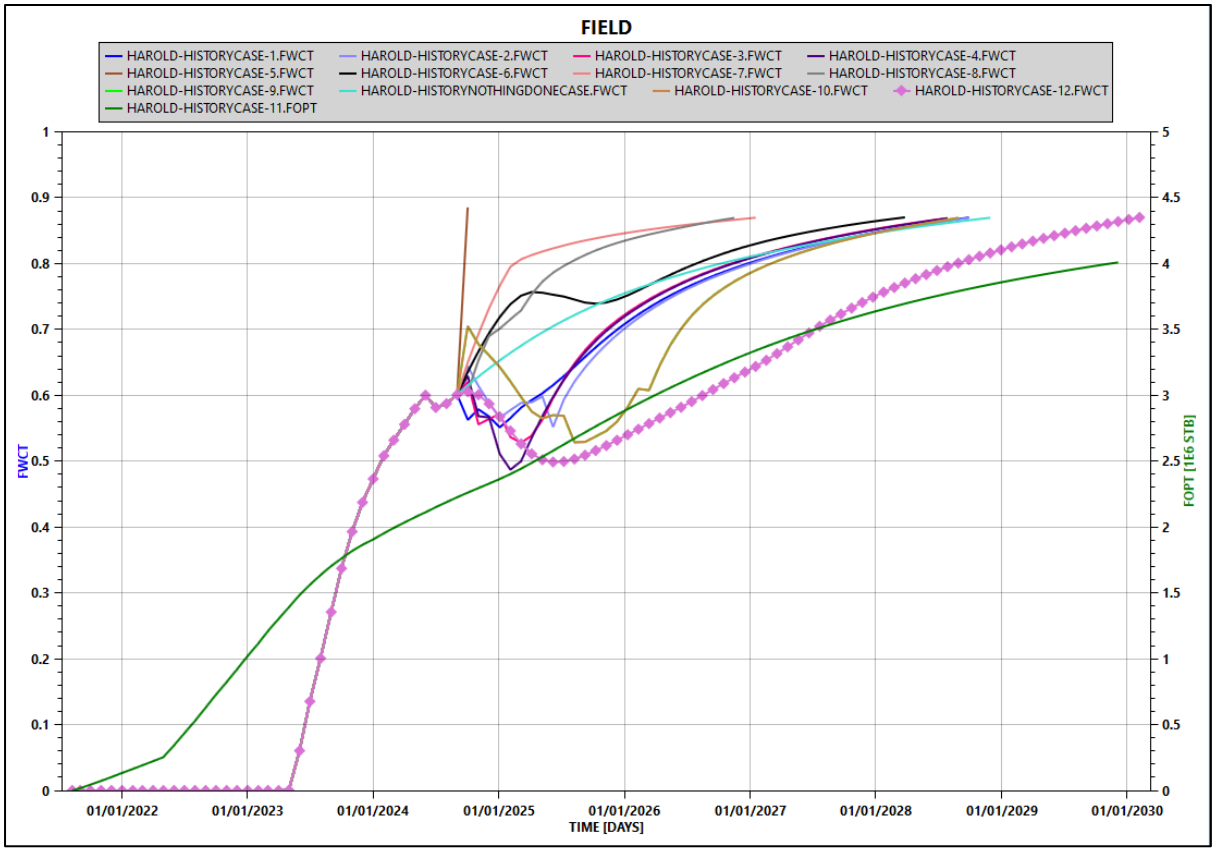


FIGURE 11 SHOWING WATER CUT FOR ALL THE CASES DONE

SATURATION PLOTS FOR THE CASES EXPLAINED IN THE REPORT

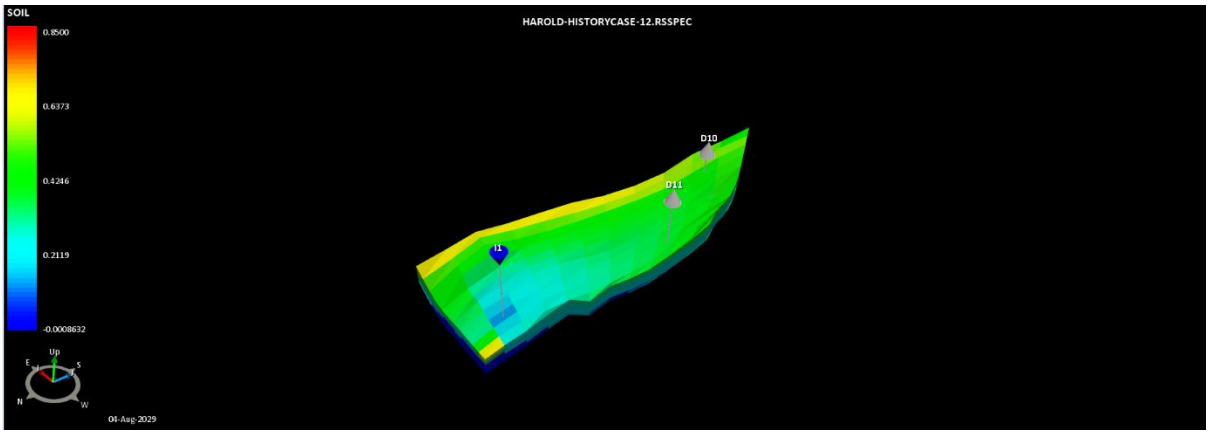


FIGURE 12 SHOWING SATURATION PLOT OF THE OPTIMUM MODEL

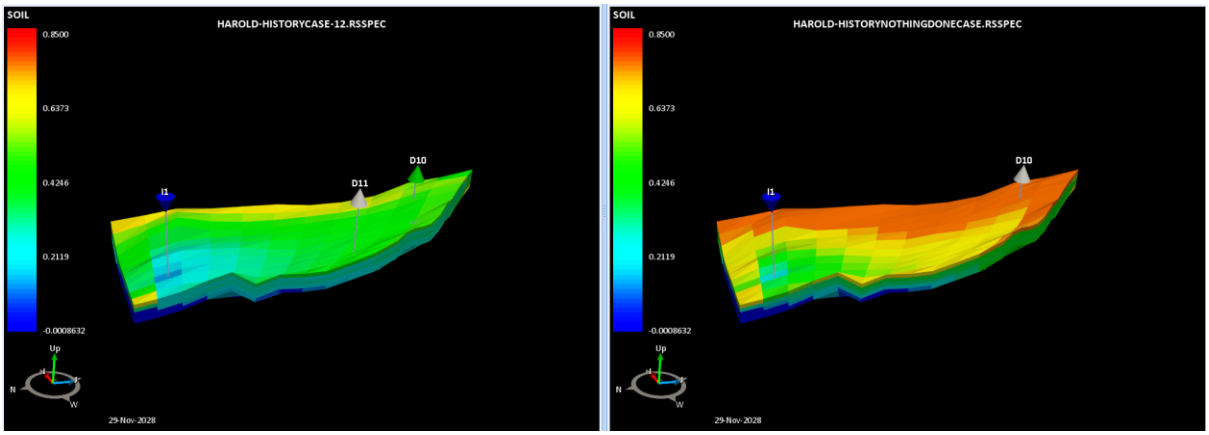


FIGURE 13 SHOWING SATURATION PLOT OF OPTIMUM MODEL AND NOTHING DONE CASE MODELS

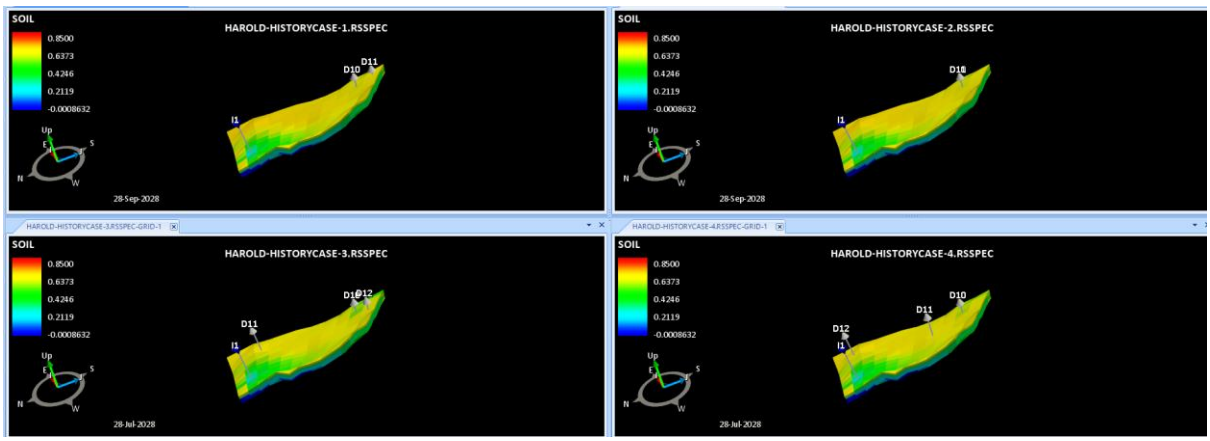


FIGURE 14 SHOWING SATURATION PLOTS FOR NEW DRIILED WELLS MODELS

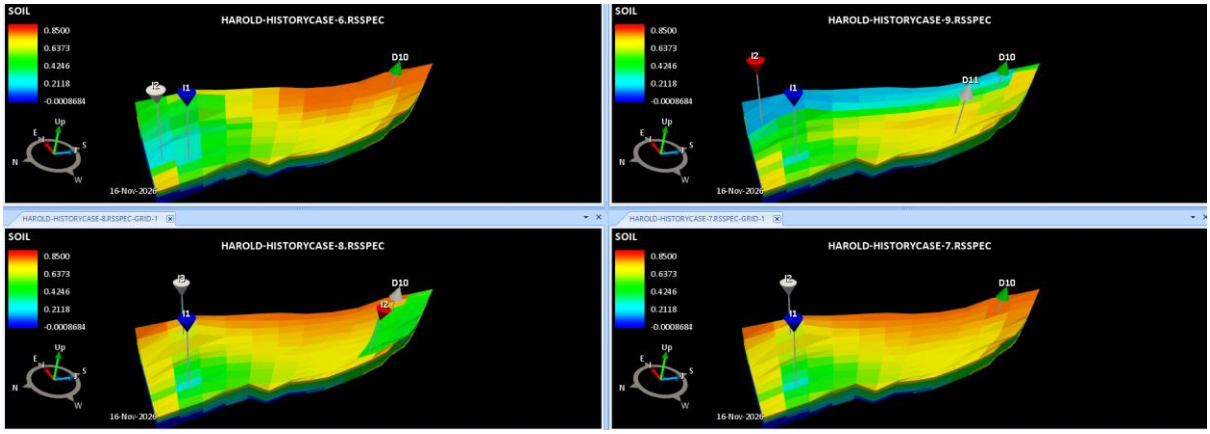


FIGURE 15 SHOWING THE SATURATION PLOT OF NEW WATER AND GAS WELLS MODELS

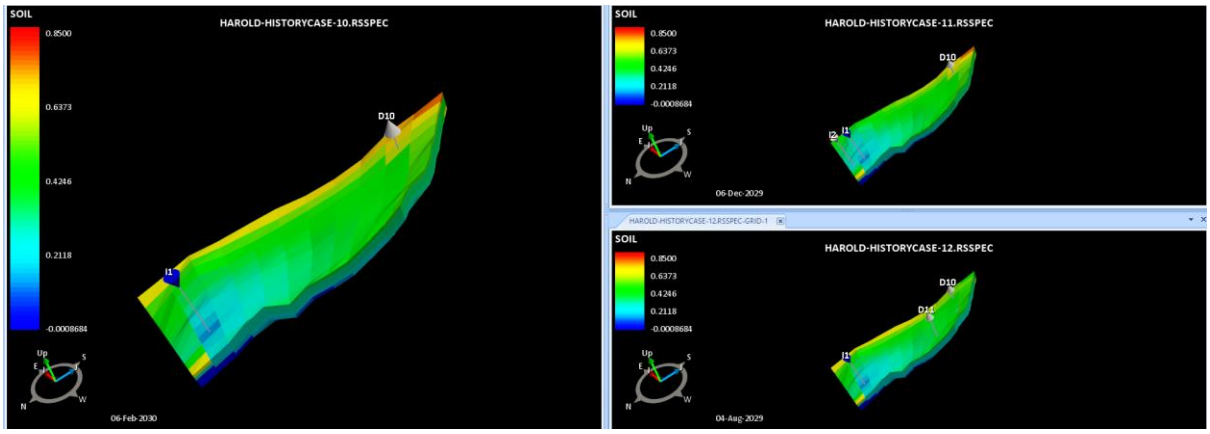


FIGURE 16 SHOWING POLYMER CASES MODELS

TABLE SHOWING SIMULATIONS MADE

SCENARIO	FILE NAME	PARAMETERS VARIED	RESULTS			
			FOE(%)	FOPT(1E6 STB)	FWCT(%)	TIME(END OF PRODUCTION)
	CASENON-DONE	NOTHING DONE CASE	0.4793	3.2442	0.8697	29/11/2028
	CASE-1	ONE VERTICAL PRODUCER WELL	0.5339	3.6142	0.86999	28/08/2028
	CASE-2	ONE HORIZONTAL PRODUCER WELL	0.5336	3.6123	0.8695	28/09/2028
DRILL NEW WELLS	CASE-3	TWO VERTICAL PRODUCER WELLS	0.5350	3.6212	0.8691	28/07/2028
	CASE-4	VERTICAL AND HORIZONTAL PROD.WELLS	0.5340	3.6146	0.8686	28/07/2028
	CASE-5	SIDETRACK	0.3306	2.2376	0.8853	02/10/2024
GAS/ WATER INJECTION	CASE-6	GAS INJECTOR WELL	0.4514	3.0556	0.8702	26/03/2028
	CASE-7	WATER INJECTION WELL	0.3972	2.6889	0.8697	17/01/2027
	CASE-8	GAS AND new WATER WELL	0.4031	2.7284	0.8693	16/11/2026
	CASE-9	HORIZONTAL WELL AND GAS INJECTION	0.5768	3.9043	0.86941	28/08/2028
	CASE-10	POLYMER FLOODING	0.6073	4.1111	0.8701	06/02/2030
POLYMER FLOODING	CASE-11	POLYMER AND GAS WELL	0.5947	4.0255	0.8692	06/12/2029
	CASE-12	POLYMER AND HORIZONTAL PROD. WELL	0.64046	4.3354	0.8692	04/08/2029